

Part A:

Question 1:

Algorithm:

- Step 1: Start
- Step 2: Take a number N from the user
- Step 3: Set Counter = N
- Step 4: Print counter
- Step 5: Counter = Counter - 1
- Step 6: If counter is greater than or equal to 1, go back to step 4
- Step 7: End

Pseudocode:

- Begin

Input N

Counter = N

~~While Counter ≥ 1 do~~ Repeat

~~Print Counter~~

Print Counter

~~Counter = Counter - 1~~

Counter = Counter - 1

End While

Until Counter ≥ 1

- End

Question 2:

Algorithm:

- Step 1: Start

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- Step 2: Take the input Y from the user
- Step 3: Check condition 1 that y is divisible by 4
- Step 4: If y is ^{not} divisible by 4, go to step ~~10~~ 11
- Step 5: If y is ~~not~~ divisible by 4, check ^{condition 2} if it's divisible by 100
- Step 6: If y is not divisible by 100, go to step 10
- Step 7: If y is divisible by 100, check condition 3 ~~that~~ that it is divisible by 400 or not.
- Step 8: If it is divisible by 400, go to step 10
- Step 9: If it is not divisible by 400, go to step 11
- Step 10: Display "It's a leap year."
- Step 11: Display "It's not a leap year"
- Step 12: End

Pseudocode:

- Begin
- ~~Input Y~~ ^{Print} Display "Enter the years"
- Input Y
- If $Y \% 4 \neq 0$ then
- If $Y \% 100 \neq 0$
- If $((Y \% 4 == 0 \text{ AND } Y \% 100 \neq 0))$

The pseudocode is written on next page.

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- Begin
-   Print "Enter the Year:"
-   Input Y
-   If ( $Y \% 4 == 0$  AND ( $Y \% 100 != 0$  OR  $Y \% 400 == 0$ )) then
       print "It's a leap year"
    else
       print "It's not a leap year"
-   Endif
- End
```

Question 3:

Algorithm:

- 1) Start
- 2) Take the input N from the user
- 3) Set $N = |N|$
- 4) Set Count = 0
- 5) If ($n > 0$), divide n by 10 and increase the count by 1
- 6) Repeat Step 5 until $n = 0$
- 7) If ($\text{count} == 1$), state it's a single digit number.
- 8) If ($\text{count} == 2$), state it's a double digit number
- 9) If ($\text{count} == 3$), state it's a triple digit number.
- 10) If ($\text{count} > 3$) state it's a more than triple digit number
- 11) Display the digit count and the classification.
- 12) End

Pseudocode:

```
- Begin
```

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- Print 'Enter the number:'
- Input y
-  $N = |N|$ 
- Count = 0
- Repeat
-      $N = N/10$ 
-     Count = Count + 1
- Until  $N == 0$ 
- If (Count == 1) then
-     Print "It's a single digit number"
- else if (Count == 2) then
-     Print "It's a double digit number"
- else if (Count == 3) then
-     Print "It's a triple digit number"
- else
-     Print "It's a more than triple digit number."
- End if
- End

```

Question 4:

Algorithm:

- 1) Start
- 2) Set table number, $n = 1$
- 3) Set multiplier, $m = 1$
- 4) If $n \bmod 3$ is equal to 0 , then add 1 to n
- 5) If $m > 10$, then add 1 to n
- 6) Multiply $m \times n$ and set the result as p

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- 7) Display $n \times m = p$
- 8) $m = m + 1$
- 9) Repeat the ~~step~~ ^{number} step 6, 7, 8 until the ~~multiplication~~ ^{number} > 10
- 10) End

Pseudocode:

- 1) Start
- 2) Set $n = 1$
- 3) Set $m = 1$
- 4) If ~~n mod 3~~ $(n \% 3 = 0)$, then
 - $n = n + 1$
 - End if
- If $(m > 10)$, then
 - $n = n + 1$
 - End if
 - Repeat
 - $p = m \times n$ ~~and not multiplication~~
 - Print " $n \times m = p$ "
 - $m = m + 1$
 - Until $n > 10$
- End

Question 5:

Algorithm:

- 1) Start
- 2) Take the input N from the user
- 3) Find the remainder of n by $n \bmod 10$ and add it to the recursive digit sum of the remaining digits

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- 4) ~~Repeat the step 3 until the number is a single digit.~~
4) Display the calculated sum as an intermediate result
5) Repeat the step 3 and 4, until the number is a single digit
6) End

Pseudocode:

- ~~Begin~~
- ~~Print "Enter the numbers"~~
- ~~Input N~~
- ~~Repeat Set sum = 0~~
- ~~Repeat~~
 - ~~Sum += ~~the digit~~ N % 10~~
 - ~~num /= 10~~
- ~~Until sum < 10~~
- ~~End~~

- Begin
- Print "Enter the numbers:"
- Input N
- Repeat
- Set Sum = 0
- Repeat
- Sum = Sum + (N % 10)
- N = N / 10
- Until N equals to 0
- Print "Intermediate sum:"
- N = Sum
- Until N < 10
- End

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